Transformations Between Deterministic and Stochastic Reinforcement-Learning Frameworks

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Abstract

Alexander and Pedersen define a transformation of reinforcement-learning frameworks by preservation of pairwise agent rankings in each transformed environment, together with two existential nontriviality conditions. We classify transformations between their four standard deterministic and stochastic frameworks under that definition. With at least two actions, a transformation exists exactly when the source agents are deterministic or the target agents are stochastic. The environment type does not affect existence. Positive cases follow from a one-parameter policy family and a three-valued choice of source environments. Negative cases follow from a finite scripted environment separating deterministic policies and the convexity of values attainable by stochastic policies. We also exhibit finite abstract frameworks showing that the full transformation notion need not compose, and distinguish it from a stronger cross-environment condition. The maps in the classification may be noncomputable and noninjective; the theorem concerns the stated mathematical definition, not effective conversion of arbitrary learned policies.

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1 What should a change of reinforcement-learning framework preserve?

A reinforcement-learning framework specifies agents, environments and the value obtained when an agent interacts with an environment. Different formalizations permit deterministic or randomized choices, different reward sets, and different conventions for who acts first. Whether such changes preserve an agent's relative ability is a natural foundational question.

Alexander and Pedersen [1] propose an abstract definition based on rankings. An agent map converts source agents into target agents; an environment map converts target environments back into source environments. Two converted agents must have the same strict ordering in a target environment as their source versions have in its converted source environment. This definition compares agents *within one environment*. It makes no demand about comparing one agent's scores across different environments.

We give a complete classification for the four standard frameworks in version 5 of their paper. A previous counterexample note [2] supplies one positive instance contradicting the claimed deterministic-to-stochastic obstruction. Here a single positive construction decides every positive case, and a separate obstruction decides every remaining case. Both arguments are included in full.

Theorem 1.1 (Classification)**.** *Assume the finite action alphabet has at least two elements. Under Definition 2.1, a transformation between the four frameworks of Definition 2.2 exists if and only if*

$$\textit{the source agents are deterministic} \quad \textit{or} \quad \textit{the target agents are stochastic.}$$

For a singleton action alphabet, no such transformation exists, including from a framework to itself.

The singleton qualification comes from the definition's nontriviality requirement, not from a technical failure of the proof. With only one policy, no pair of image agents can have different values.

2 Frameworks, interactions and the exact transformation condition

An abstract framework is a triple $F = (A, E, V)$ with $V : A \times E \to \mathbb{R}$. Agents and environments need not have effective descriptions at this level.

Definition 2.1. A transformation from $F = (A, E, V)$ to $\overline{F} = (\overline{A}, \overline{E}, \overline{V})$ is a pair of maps $f : A \to \overline{A}$, $g : \overline{E} \to E$ such that

$$\overline{V}(f(p), \mu) < \overline{V}(f(q), \mu) \quad \Longleftrightarrow \quad V(p, g(\mu)) < V(q, g(\mu)) \tag{2.1}$$

for every p, q, μ, and such that:

1. some two image agents have strictly different values in one target environment;
2. some image agent has strictly different values in two target environments.

Equation (2.1) also preserves ties: apply it in both orders and use the trichotomy of real order. It does not require f or g to be injective, computable, continuous, or faithful to numerical values themselves. In particular the two nontriviality clauses are existential requirements on the image, not global information-preservation conditions.

2.1 The four standard frameworks

Fix finite disjoint alphabets $\mathcal{A}$ of actions and $\mathcal{P}$ of percepts. Each is nonempty. A reward function $r : \mathcal{P} \to \mathbb{Z}$ has both 0 and 1 in its range. Choose percepts z_0, z_1 with those rewards. Interaction is percept-first:

$$z^{(1)}, a^{(1)}, z^{(2)}, a^{(2)}, \ldots.$$

An agent history is a finite alternating sequence ending in a percept; an environment history is empty or ends in an action. A deterministic policy gives one action at every agent history. A stochastic policy gives a rational probability distribution on actions at every such history. Environments similarly give a percept or a rational distribution on percepts. A deterministic object is identified with its point-mass stochastic version when necessary.

Policies are defined on all histories of the appropriate form, including histories which would have probability zero in a particular interaction. They have no effectiveness restriction.

Let $V_n(p, \mu)$ be the expected sum of rewards in the first n percepts of the alternating process. An environment is *admissible* when $V(p, \mu) = \lim_n V_n(p, \mu)$ exists as a finite real for every policy of either kind. There is no common normalization such as $|V| \leq 1$. This condition is stronger than existence of a value just against the policies occurring in the image of a proposed transformation.

Definition 2.2. For $X, Y \in \{D, S\}$, write F_{XY} for the framework with agent type X and environment type Y, using all agents of that type and all admissible environments of that type. The first subscript always denotes agents. D means deterministic and S stochastic.

These are the four frameworks in [1, Definition 9]. The finite play convention is the alternating percept-first process stated there; values here are defined directly by that process.

3 The positive cases: one scalar of agent behavior is enough

Choose distinct actions a_0, a_1. For a source policy p, define

$$t(p) = p(a_0 \mid \langle z_0 \rangle).$$

For a deterministic source, interpret this as 1 if its first response to z_0 is a_0, and 0 otherwise. Thus t is binary in that case; for stochastic sources it is rational in $[0, 1]$. Both endpoints are attained by constant-action policies.

For each rational $t \in [0, 1]$, define a target policy Q_t which, at its first action, chooses a_1 with probability t and a_0 with probability $1 - t$, independently of the initial percept. At every later history it chooses a_0. This defines the policy on every history; no persistent hidden random seed is required. The endpoints Q_0, Q_1 are deterministic.

Lemma 3.1. *For every admissible target environment μ,*

$$V(Q_t, \mu) = A_\mu + t(B_\mu - A_\mu), \qquad A_\mu = V(Q_0, \mu),\ B_\mu = V(Q_1, \mu).$$

Proof. For each finite horizon, the trace law of Q_t is the mixture of the trace laws of Q_0 and Q_1 with weights $1-t, t$. The only varying policy choice is the first action; after it all three policies have the same prescribed continuation. Conditioning on the first percept does not alter those weights. The environment may depend on the whole history, but for a fixed trace its own probability factors are the same in all three comparisons. Finite summation gives the identity for V_n. Take the limit of this two-term linear combination. Both endpoint limits are finite by admissibility. □

Three simple source environments suffice to match the possible orderings in this affine family. Each emits z_0 initially and zero rewards forever after its second percept:

- e_+ gives reward 1 at the second percept exactly when the first action was a_0;
- e_- gives reward 1 exactly when that first action was not a_0;
- e_0 always gives reward zero.

Choose z_1 or z_0 to implement the reward. These rules specify the environments on off-path histories as well. Their values against any source policy are $t(p), 1-t(p), 0$, respectively. They are deterministic and admissible against every deterministic or stochastic policy.

Proposition 3.2. *If source agents are deterministic or target agents are stochastic, there is a transformation.*

Proof. Set $f(p) = Q_{t(p)}$. Its codomain has the required type: either only the deterministic endpoints are needed or all rational Q_t are allowed. For any target environment put

$$g(\mu) = \begin{cases} e_+, & B_\mu > A_\mu, \\ e_-, & B_\mu < A_\mu, \\ e_0, & B_\mu = A_\mu. \end{cases}$$

Use the point-mass version if the source environment class is stochastic. The affine identity shows that every image pair is ordered by increasing t, decreasing t, or a constant value according to the same three cases. The source environments realize exactly these orderings, so (2.1) holds for every pair, including ties, in every target environment.

For the first nontriviality condition, take a target environment which emits z_0, then rewards the first action a_1, then gives zeros. It assigns values 0 and 1 to Q_0, Q_1, both of which occur in the image. For the second, compare the always-zero target environment with one which emits z_0, then z_1 independently of action, then zeros. Every image policy has values 0 and 1 in these two environments. All these environments are admissible in either environment type. □

The sign choice in g is a set-theoretic comparison of two real values. It need not be computable from any description of μ. This is allowed by the definition, and is central to the scope of the theorem.

4 A finite environment which separates deterministic policies

The negative argument needs more than two policies being syntactically different. They can differ only at histories inconsistent with their own previous actions and still have identical value in every environment. An actual value distinction avoids that problem.

Lemma 4.1 (Two-value separator)**.** *Suppose deterministic policies α, β have different values in some admissible environment. There is an admissible deterministic environment η and an integer C such that every deterministic policy has value either C or $C+1$ in η, with α and β taking different values.*

Proof. There must be a finite prescribed percept sequence on which their recursively generated actions first differ. Otherwise their actions would agree along every prescribed percept sequence; induction on trace length would then give identical finite trace distributions in every environment, and consequently identical limiting values.

Choose a sequence $y_1, \dots, y_k$ with first action divergence at stage k. Let $b_1, \dots, b_{k-1}$ be the common earlier actions and b_k the action taken by α at stage k. Define η to emit this entire percept script, independently of actions. At percept $k+1$ emit z_1 if and only if the recorded action sequence is exactly $b_1, \dots, b_k$; otherwise emit z_0. Emit z_0 thereafter.

Every deterministic policy receives the same script reward $C = \sum_{i=1}^{k} r(y_i)$ and then either 0 or 1. In particular α receives $C+1$ and β receives C. Every stochastic policy has value between these endpoints. All nonzero rewards occur at a fixed finite horizon, so admissibility holds against the full policy class. □

It matters that the script continues after an unexpected action. Aborting it on deviation could change the script reward for third policies and destroy the two-value conclusion. The offset C also allows prescribed percepts to carry arbitrary integer rewards; no zero-reward copy of every observation is assumed.

5 Mixing policies without a hidden random seed

For stochastic source policies we need a policy whose value is the midpoint of two given values in *every* admissible environment. Pointwise averaging of action distributions is generally insufficient: it can redraw the branch at each step instead of mixing entire plays. The correct behavioral policy updates the branch weights from the recorded action history.

For an agent history h, let $L_p(h)$ be the product of the probabilities that p assigns to the earlier actions recorded in h, each evaluated at its recorded preceding history. Do not multiply probabilities of percepts. Let $0 < w < 1$ be rational. Define

$$m(a \mid h) = \frac{wL_p(h)p(a \mid h) + (1-w)L_q(h)q(a \mid h)}{wL_p(h) + (1-w)L_q(h)} \tag{5.1}$$

when the denominator is positive; otherwise use the point mass at a_0. This is a rational distribution at every history.

Lemma 5.1. *The policy m satisfies*

$$V(m, \nu) = wV(p, \nu) + (1-w)V(q, \nu)$$

for every admissible environment ν.

Proof. Induction over recorded actions gives $L_m(h) = wL_p(h) + (1-w)L_q(h)$. For a positive denominator, multiplication by (5.1) cancels that denominator and gives the identity at the next action. For a zero denominator both nonnegative summands vanish, as do all their action extensions; the arbitrary off-path choice cannot revive them. Extending a history by a percept leaves the action-only products unchanged.

For each finite full trace in a fixed environment, its percept-likelihood product is a common factor multiplying these three action products. It may depend on the recorded actions; that does not matter because the trace is fixed in this comparison. Finite trace probabilities, and then finite reward expectations, satisfy the same mixture identity. Take limits of the two-term linear combination. Admissibility supplies the required finite limits for all three policies. □

6 The negative cases and the complete table

Proposition 6.1. *There is no transformation with stochastic source agents and deterministic target agents, for either environment type.*

Proof. Suppose (f, g) were one. The first nontriviality condition supplies source policies p, q whose deterministic images have different values in some target environment. Lemma 4.1 supplies a deterministic environment η giving every deterministic target policy one of two values $C, C+1$, with $f(p), f(q)$ taking opposite values. Its point-mass version is also available in a stochastic target environment class.

Faithfulness makes $V(p, g(\eta))$ and $V(q, g(\eta))$ distinct. Let m be their midpoint policy from Lemma 5.1, with $w = 1/2$. Its source value is strictly between them. Applying (2.1) to the two comparisons forces $V(f(m), \eta)$ strictly between C and $C+1$. No deterministic target policy has such a value in η. This is a contradiction. □

The two propositions prove Theorem 1.1 when there are at least two actions. If there is one action, both policy classes have only one member, so the first nontriviality condition fails for every map. The full table is therefore, for at least two actions,

source\target	DD	DS	SD	SS
DD	yes	yes	yes	yes
DS	yes	yes	yes	yes
SD	no	no	yes	yes
SS	no	no	yes	yes

The environment type disappears from the criterion. Under this definition there are two mutual-transformability classes, separated by agent randomization, and a one-way transformation from the deterministic class to the stochastic class. This concerns the specific four frameworks; it is not a general composition theorem.

7 Composition and a stronger comparison condition

7.1 The existential nontriviality conditions need not survive

Faithfulness alone composes: substitute one pair of maps into the other in (2.1). The full definition has an additional issue: witnesses to nontriviality can fall outside the image of the composite agent map or outside the selected environments.

Proposition 7.1. *There are transformations $F \to G$ and $G \to H$ whose naturally composed maps fail both nontriviality conditions.*

Proof. The following value matrices give an example, with agents as rows and environments as columns:

$$V = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 2 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 2 \end{pmatrix}.$$

From V to W, map the two agents to rows 1 and 2, and map the three target columns to source columns 1,2,2. On the selected rows the entries agree exactly, so faithfulness holds. Column 1 separates the image rows; image row 2 varies between columns 1 and 2. From W to Z, use the identity row map and select source columns 2,3. Again entries agree exactly. Row 3 witnesses both nontriviality conditions. The composite selects the first two rows of Z, all of whose entries are zero, so neither condition holds.

□

This example addresses composition of the given maps. It does not exclude the existence of a different transformation from F to H.

7.2 Comparing scores across environments changes the problem

A stronger requirement would ask, for all p, q, μ, ν,

$$\overline{V}(f(p), \mu) < \overline{V}(f(q), \nu) \quad \iff \quad V(p, g(\mu)) < V(q, g(\nu)). \tag{7.1}$$

Definition 2.1 is the special case $\mu = \nu$ and does not imply this condition. The difference has mathematical consequences.

Proposition 7.2. *Assume source agents and environments are deterministic, with rewards in a fixed lattice $d^{-1}\mathbb{Z}$, $d \geq 1$, and convergent total rewards. Assume the target has rational initial mixtures of any two admissible environments. There is no pair satisfying* (7.1) *and the second nontriviality condition.*

Proof. A convergent sequence of source partial reward sums in $d^{-1}\mathbb{Z}$ is eventually constant. Every source value therefore lies in that lattice. By (7.1), all target values attained by image agents have an order-preserving image in this lattice: equality is also preserved by applying strict comparison in both orders.

The second nontriviality condition gives one image agent with values $a < b$ in two target environments. Rational initial mixtures produce values $a + t(b - a)$ for all rational $0 < t < 1$. These would map to arbitrarily many distinct lattice points between the fixed lattice images of a and b, an impossibility. □

Initial mixtures can be realized behaviorally by the environment analogue of (5.1), using past percept likelihoods. For each policy, finite expectations are mixed and finite limits exist; hence these environments remain admissible. Finite rational reward alphabets have a common denominator and satisfy the same lattice premise. An unrestricted rational reward alphabet does not.

The proposition explains why an argument based on ordering one agent across several environments cannot establish an impossibility under the weaker printed definition. It establishes an obstruction for a different contract. It does not classify every case under that stronger contract.

8 Interpretation and further questions

The positive maps retain a single first-action probability and may collapse almost all other behavior. That is enough because the target environment map need preserve only one environment's ordering at a time. The negative argument identifies a feature that this collapse cannot erase: a nontrivial stochastic source has intermediate policy values, whereas a scripted deterministic target environment can make all deterministic values binary up to a common offset.

Consequently the classification is a result about the expressive strength of a ranking-preservation definition. It supplies no general method for compiling a trained policy into an equally capable policy in another implementation. Requiring computable maps, injectivity, preservation of actual rewards, a reward normalization, or cross-environment comparisons would each define a further problem. These changes should be stated separately before attempting another classification.

Preparation. This is an AI-assisted hand-proof manuscript. The complete four-framework argument is included, using the versioned source definition. No formal verification or established historical priority is claimed.

References

[1] S. A. Alexander and A. P. Pedersen, *Representation and Invariance in Reinforcement Learning*, arXiv:2112.07752v5, 11 August 2026. Especially Definitions 2 and 6–9 and the concrete classification section.

[2] Logic Lineages project, *A counterexample to the transformation classification as written*, working note, 2026. Public note and proof. The present manuscript includes all arguments it needs from that note.